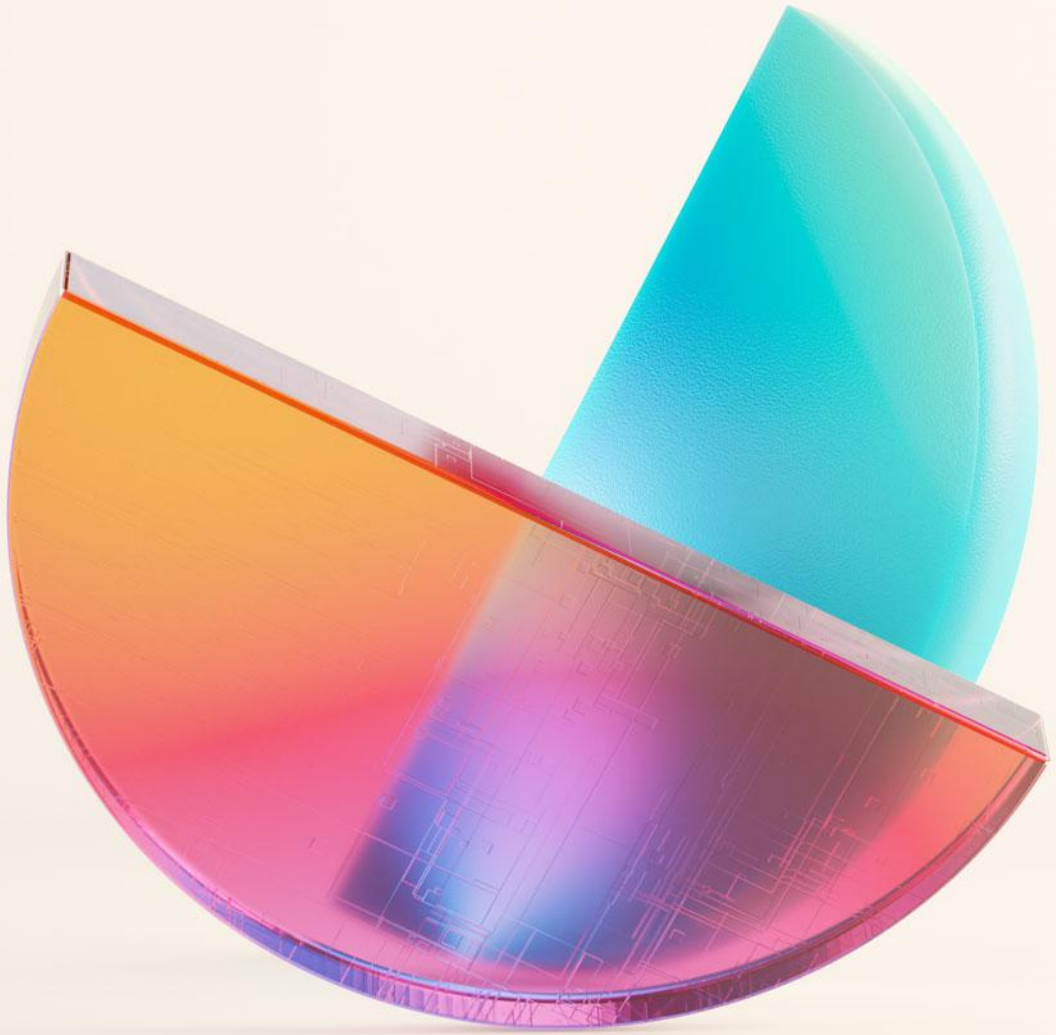
Microsoft

Microsoft Fabric
Community Conference



Building a **Fortress** of Your Fabric Environment: **Security** Best Practices for Data Engineers

Let’s connect

Erwin de Kreuk

- Principal Consultant
- Lead Data & AI InSpark

@erwindekreuk.bsky.social

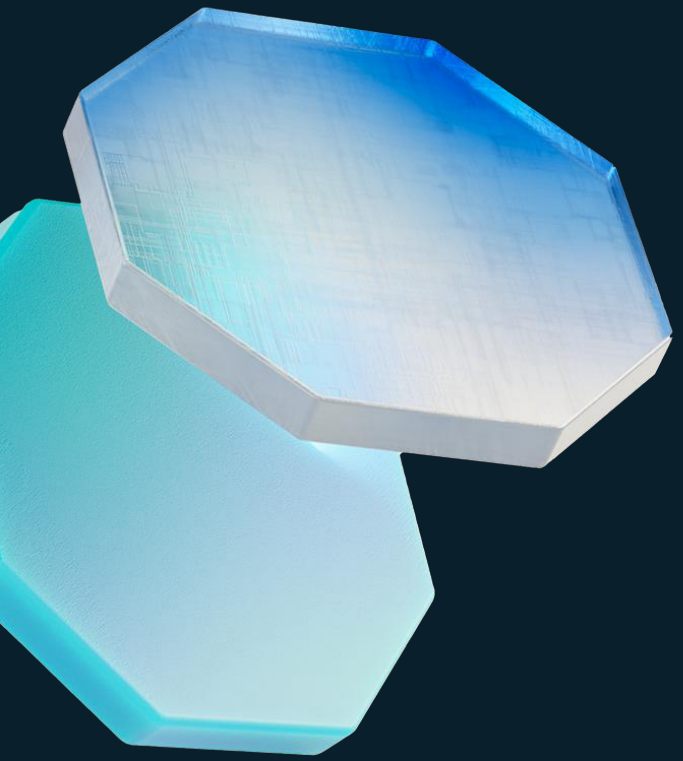
linkedin.com/in/erwindekreuk

erwindekreuk.com

github.com/edkreuk

https://sessionize.com/erwin-de-kreuk/

Dutchfabricusergroup.com





Microsoft® MVP Most Valuable Professional

Microsoft

INSPARK

Winner

Partner of the Year Awards



Country Award Data & AI

Global Award Identity





Cybercriminals are getting more creative just like us

IT Security Challenges

- **Increased Attack Surface**
- **Digital transformation accelerated the adoption of new interconnected technologies**

- **Compliance & Regulatory requirements**





- **Lack of resources**



Why data security?

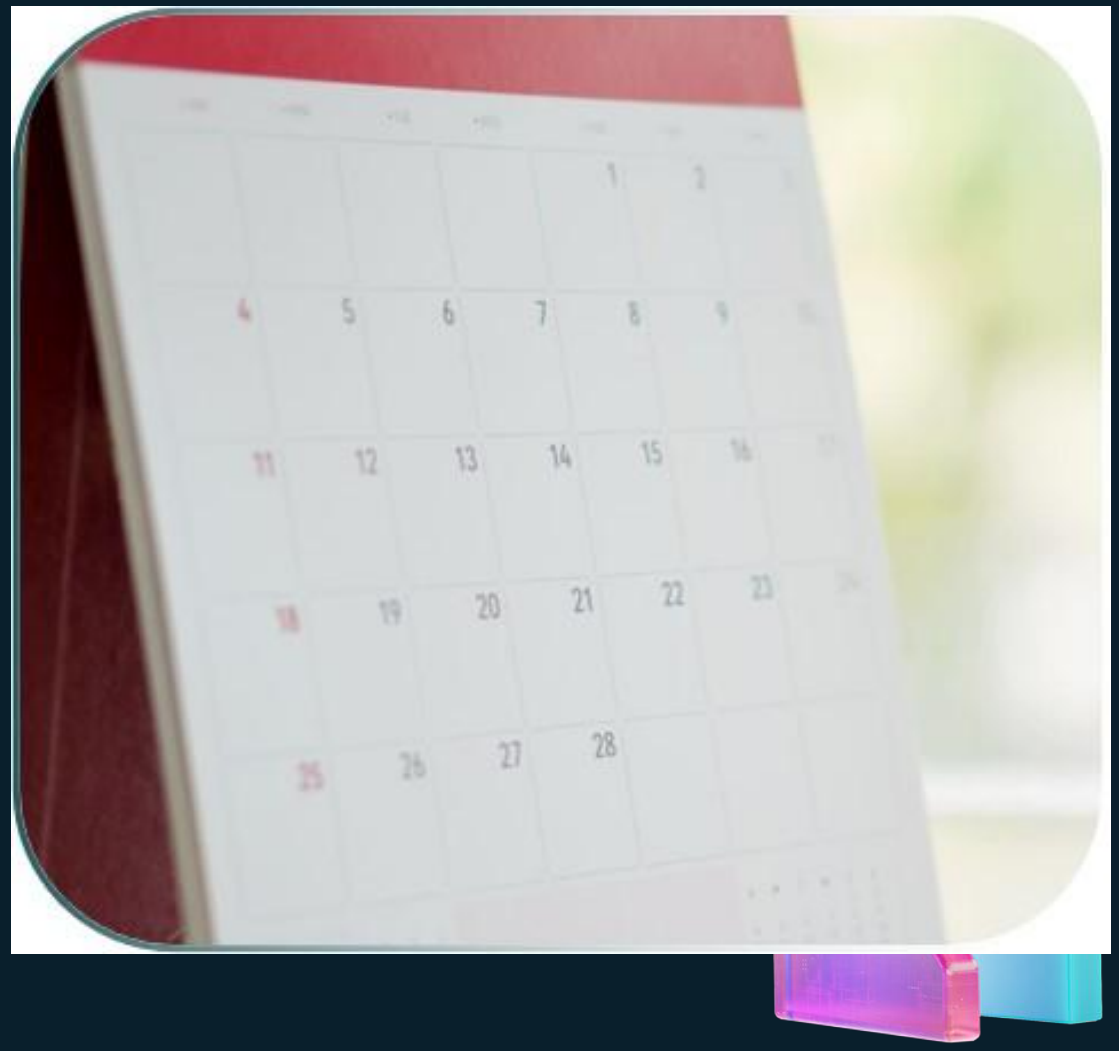
Data is most secure if no one can access it.

But…if no one can access data then you can’t use it for anything.

* The value of data | World Economic Forum (weforum.org)

Agenda

- Microsoft Fabric
- Challenges
- Entra ID
- Inbound / Outbound connections
- Microsoft Purview
- Q & A



Fabric Security Whitepaper

- Who has ever heard of the Fabric Security Whitepaper?
- Only 127 pages

- Microsoft Fabric security white paper - Microsoft Fabric | Microsoft Learn



Introduction

Security is a top priority for any organization that wants to succeed in the digital age. You need to safeguard your assets from threats and follow your organization's security policies. This whitepaper serves as an end-to-end security overview for Microsoft Fabric. It covers details on how Microsoft secures your data by default as a SaaS service and how you can secure, manage, and govern the data stored in Microsoft Fabric as an organization.

The contents of this whitepaper is created by combining several security related online documents into a single whitepaper for reading convenience. This whitepaper will be updated regularly, but the online documentation will always be up to date. You can find the online documentation here: Microsoft Fabric security - Microsoft Fabric | Microsoft Learn

A world awash with data...

In today's digital age, data is everywhere. From social media interactions to online transactions, the amount of data generated every second is staggering. This vast ocean of information holds immense potential.

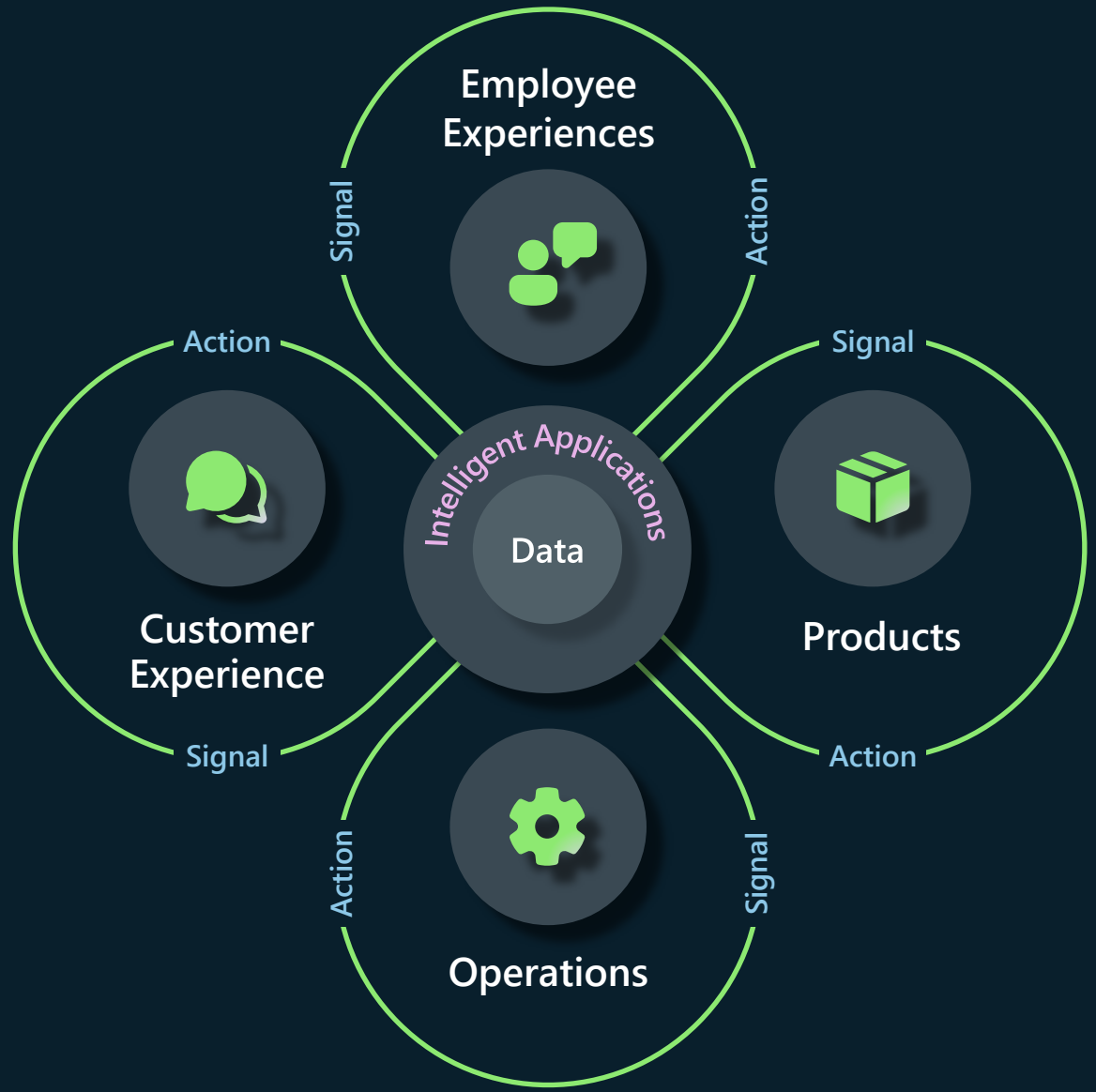
AI is changing the world

Artificial Intelligence (AI) is at the forefront of this transformation. By leveraging data, AI systems can learn, adapt, and make decisions with unprecedented accuracy and speed. From healthcare to finance, AI is revolutionizing industries, driving innovation, and improving lives.

Data is the oxygen of our digital transformation

"A new kind of company — we call them insights driven businesses — has formed. They are growing at an average of more than 30% annually"

Forrester Analytics Business Technographics Global Data & Analytics Survey



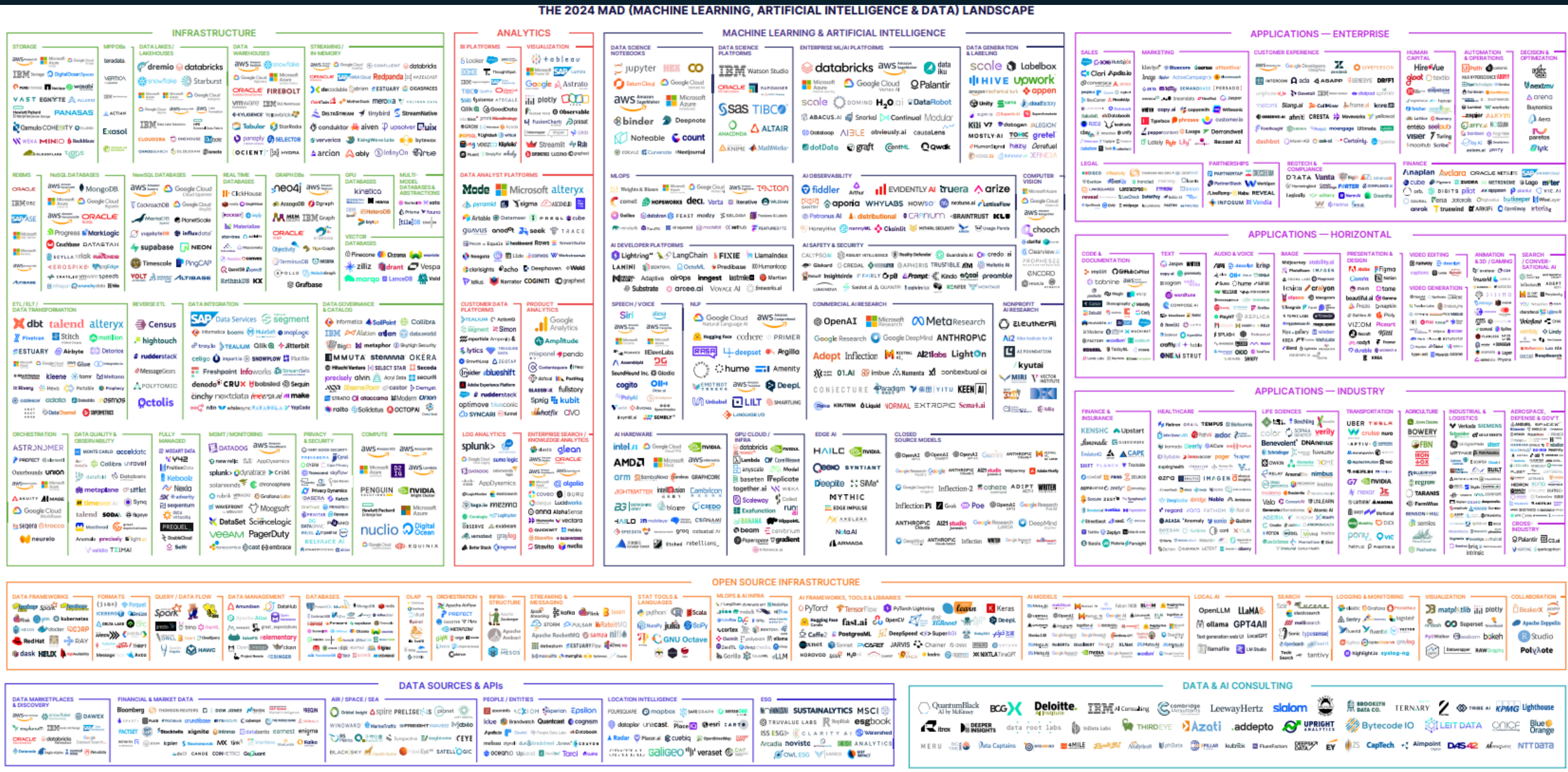
Security : The Cornerstone of Microsoft Fabric

Security is paramount in the digital age, and Microsoft Fabric ensures your data is protected at every stage. By using Fabric, organizations can trust that their data is secure, allowing them to focus on innovation and growth.

How do you translate data into competitive advantage



The 2024 ML, AI, and Data Landscape



Version 1.0 - March 2024 © Matt Turck (@mattturck) , Aman Kabeer (@AmanKabeer11) & FirstMark (@firstmarkcap) Blog post: mattturck.com/MAD2024 Interactive version: MAD.firstmarkcap.com Comments? Email MAD2024@firstmarkcap.com

FIRSTMARK
EARLY STAGE VENTURE CAPITAL

Unify,

I am the Chief Information Officer and don’t want to be the Chief Integration Officer.

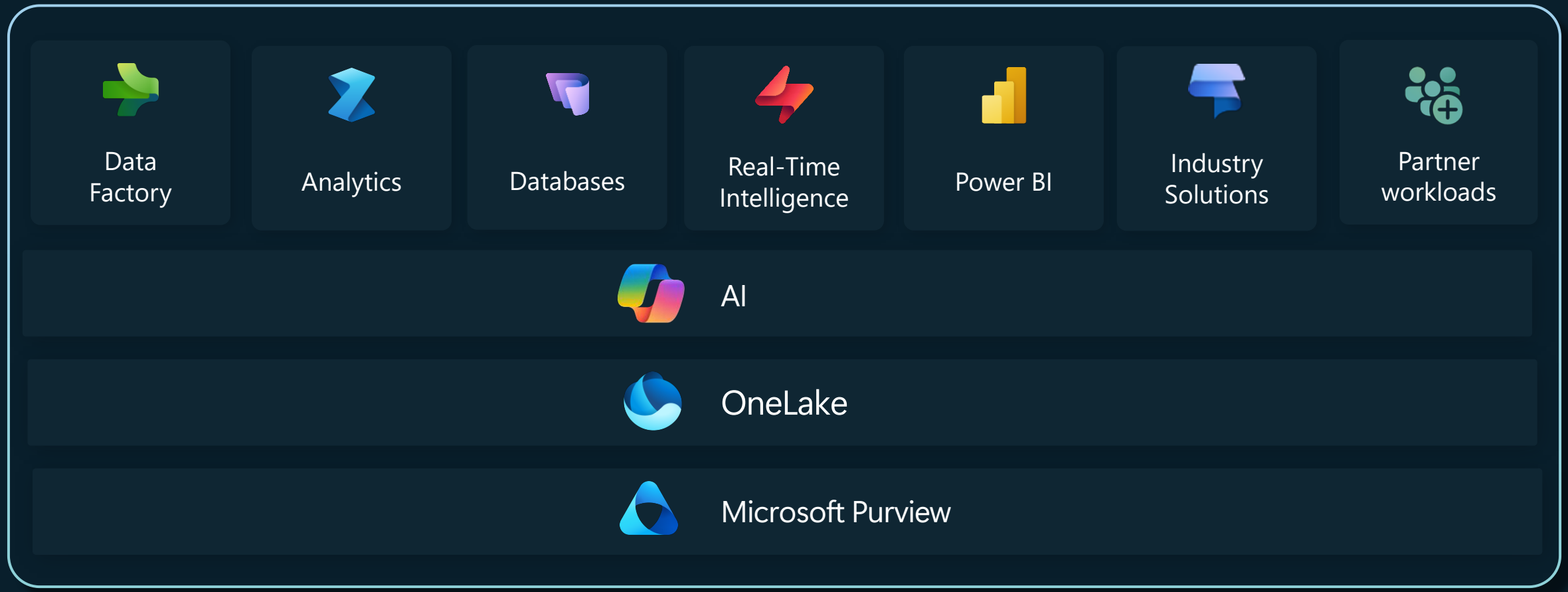
Help me govern and secure my data estate.

Every CIO, Every Enterprise



Microsoft Fabric

The unified data platform for AI transformation



Modern data challenges

- Bring data to the masses, everyone should have access to data to make decisions
- Instant access to the latest data, no copying around.
- Modern workforce, anywhere, any device.
- At the same time, you need to secure, govern and audit your data to protect customers and the company.
- SaaS platforms are designed with these challenges in mind.
- Shift from siloed PaaS to integrated SaaS.

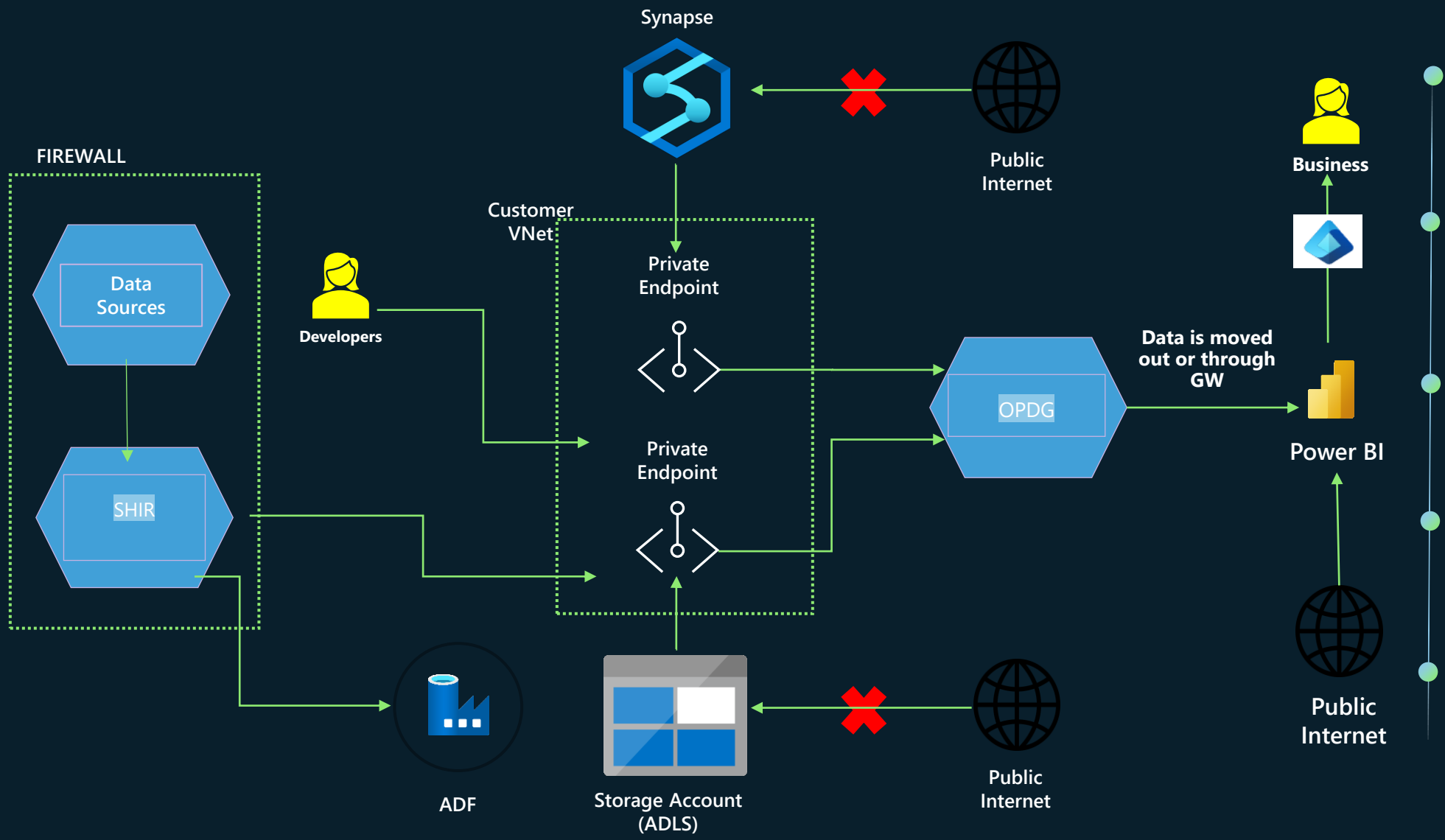


Common network security requirements

- Need to be able to connect to data inside a firewall\private link from Fabric (outbound).
- Inbound protection (restrict inbound by network location).
- More stringent customers (FSI\HLS):
  - Traffic needs to be private (not via public internet)
  - Endpoints should not be open to the public internet

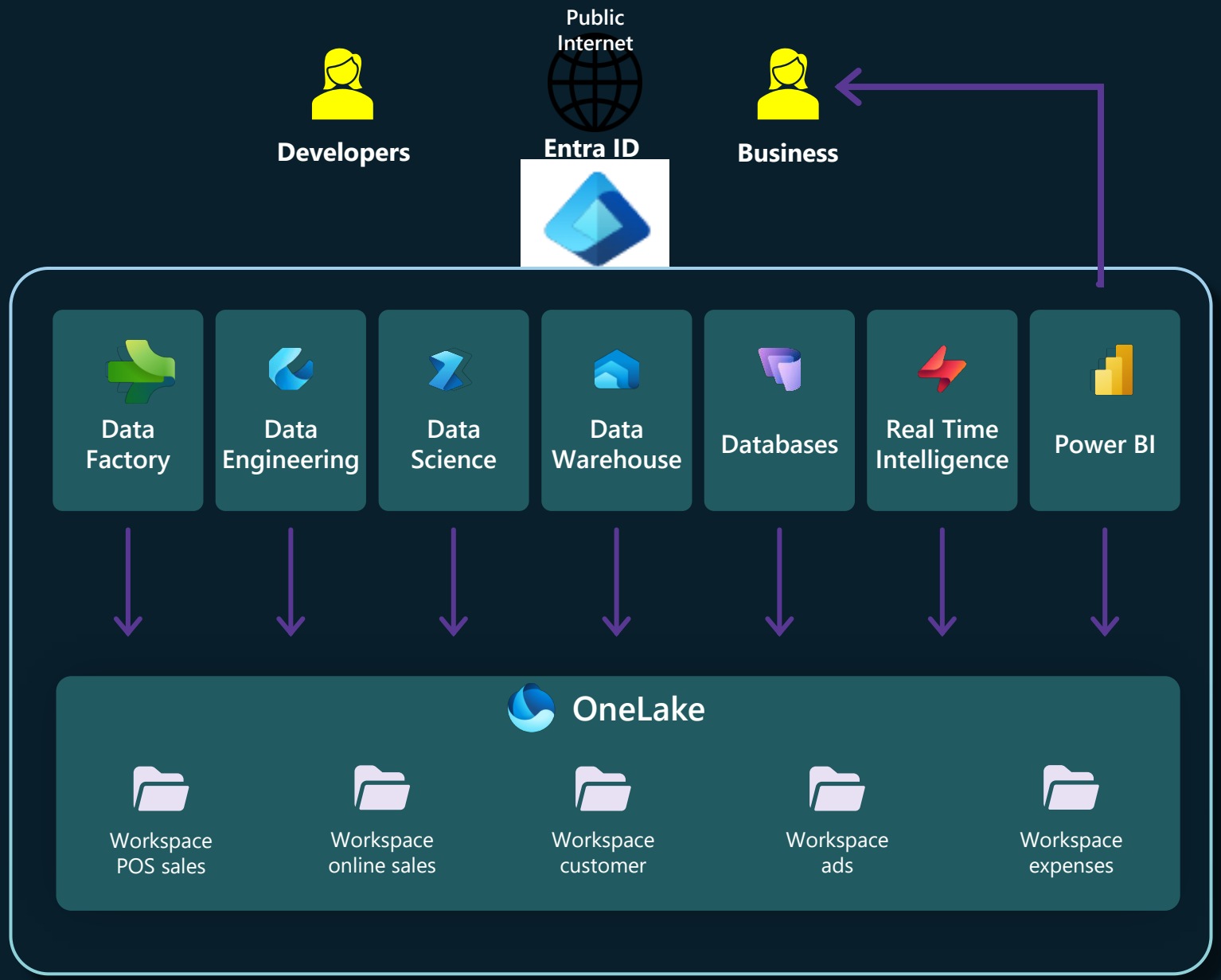


Existing PaaS World



- PaaS services are disconnected from the Public internet.
- All communications between the services are configured to go through private endpoint
- Developers connect to a VPN and connect to the backend services
- Data sources are connected through the self-hosted runtime inside the firewall
- When the CEO access the data in Power BI Report, the data is imported into Power BI or through the GW with DQ.

Microsoft Fabric – SaaS World



- Users connect to the SaaS service from global network
- Endpoints and access are protected using Entra ID
- All internal communications between the experiences happens through MS backbone network
- When the CEO access the data in Power BI Report, the data is fetched directly from the OneLake instantly and securely without copying or moving

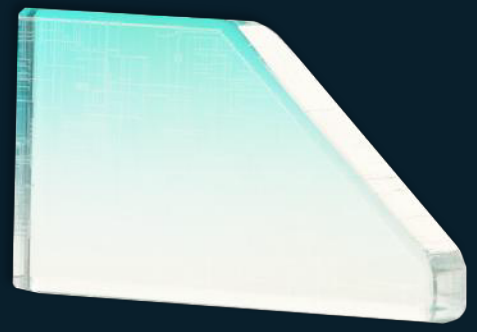
Security layers in Microsoft Fabric



- Regulations & Certification
- Data Encryption
- E2E Auditability
- High Availability & Disaster Recovery
- Microsoft Purview Information Protection
- Advanced tools in Microsoft Purview
- 3rd party and in-house governance & security solutions

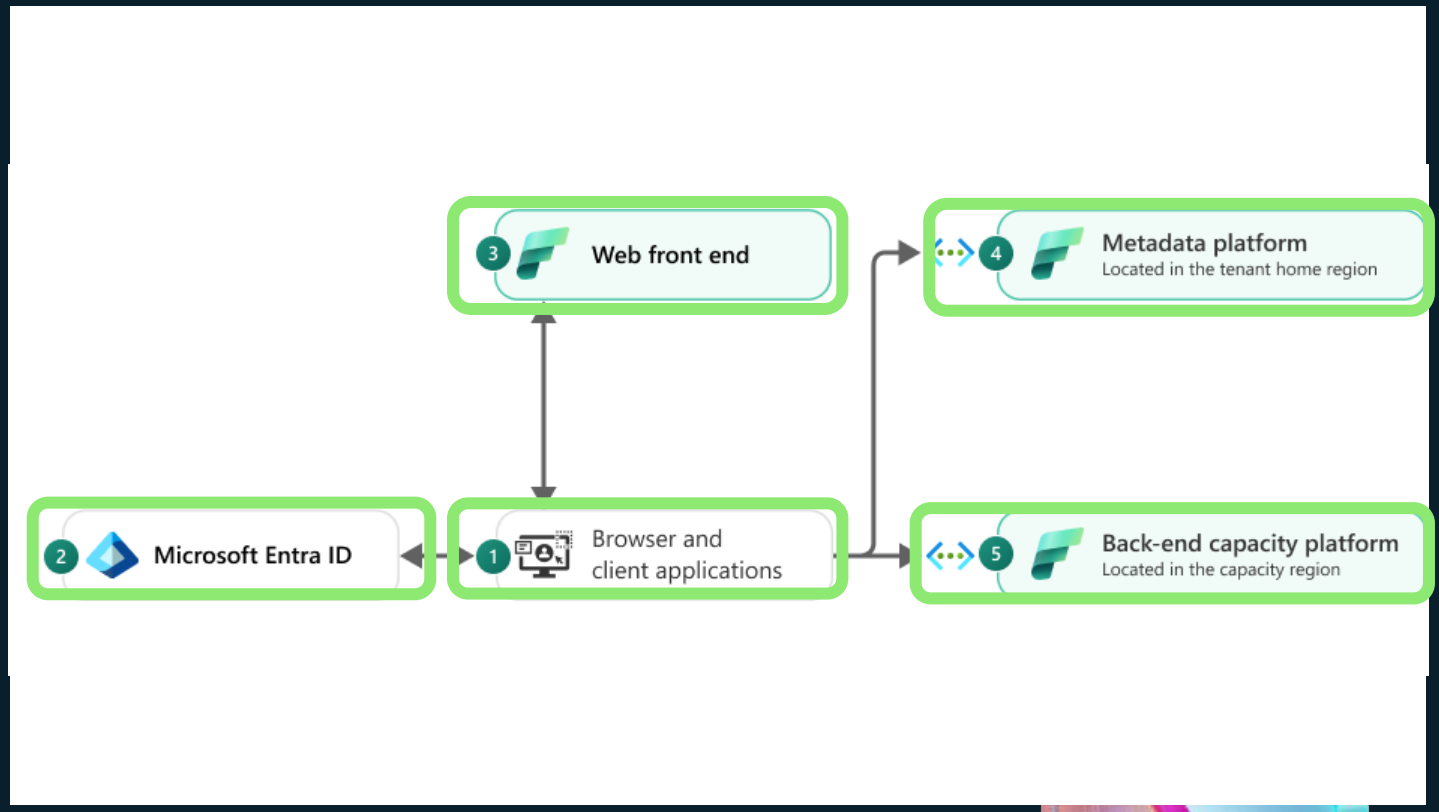
Entra ID





Microsoft Fabric Architecture

1. A user uses a browser or a client application, like Power BI Desktop, to connect to the Fabric service.
2. Authentication is handled by Microsoft Entra ID
3. The web front end receives user requests and facilitates sign-in.
4. The metadata platform stores tenant metadata, which can include customer data. Fabric services query this platform on demand in order to retrieve authorization information and to authorize and validate user requests. It's located in the tenant home region.
5. The back-end capacity platform is responsible for compute operations and for storing customer data, and it's located in the capacity region.



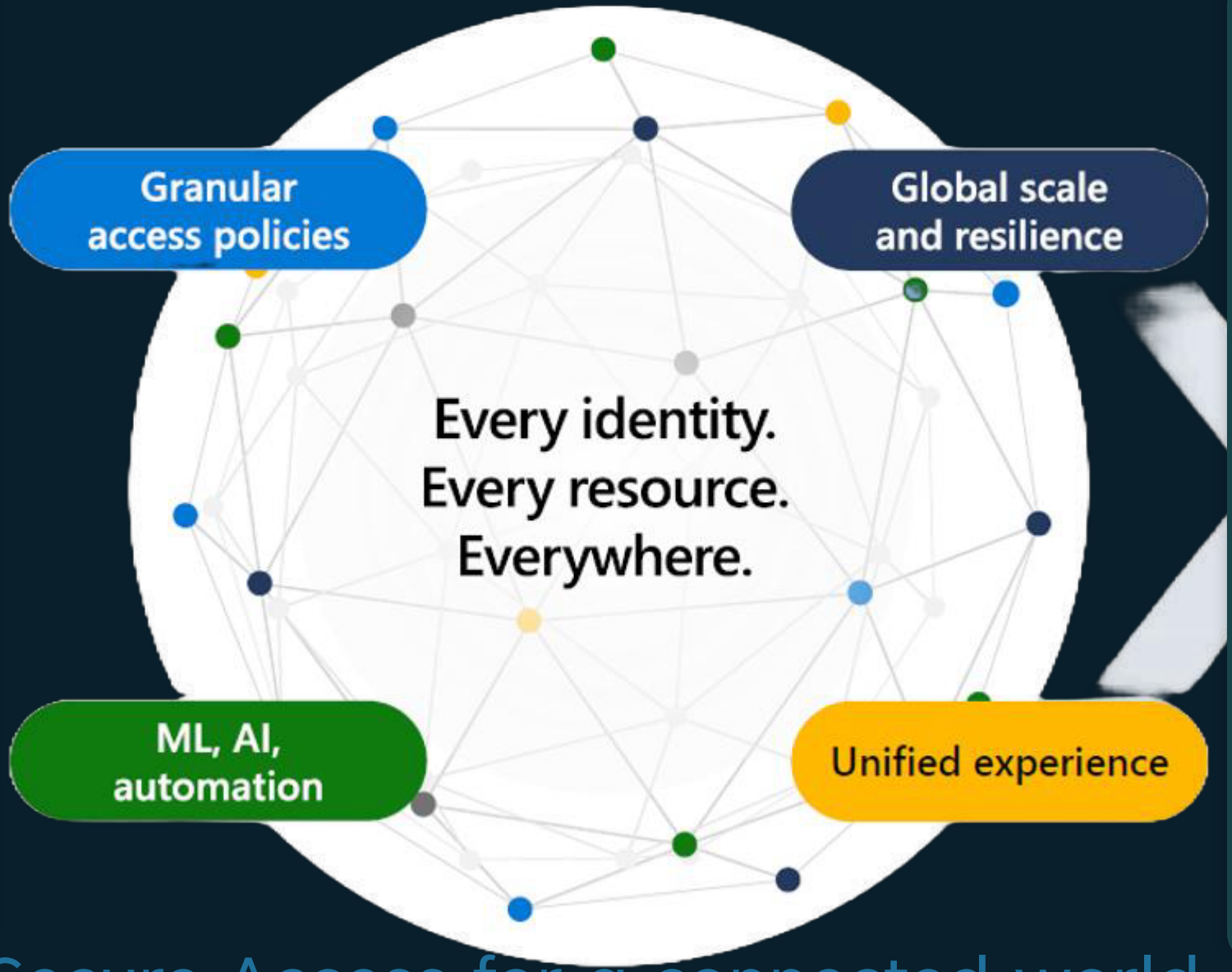
Microsoft Entra

Any employee
Cloud-based and on-premises identities, groups and roles

Any location
HQ, branch office, home, remote

Any platform
Android, iOS, Linux, MacOS, Windows

Any device
Corporate and personal



Any data, apps or resources

IaaS, PaaS, Datacenter

Microsoft 365

Microsoft Fabric

SaaS, websites

On-premises

Secure Access for a connected world



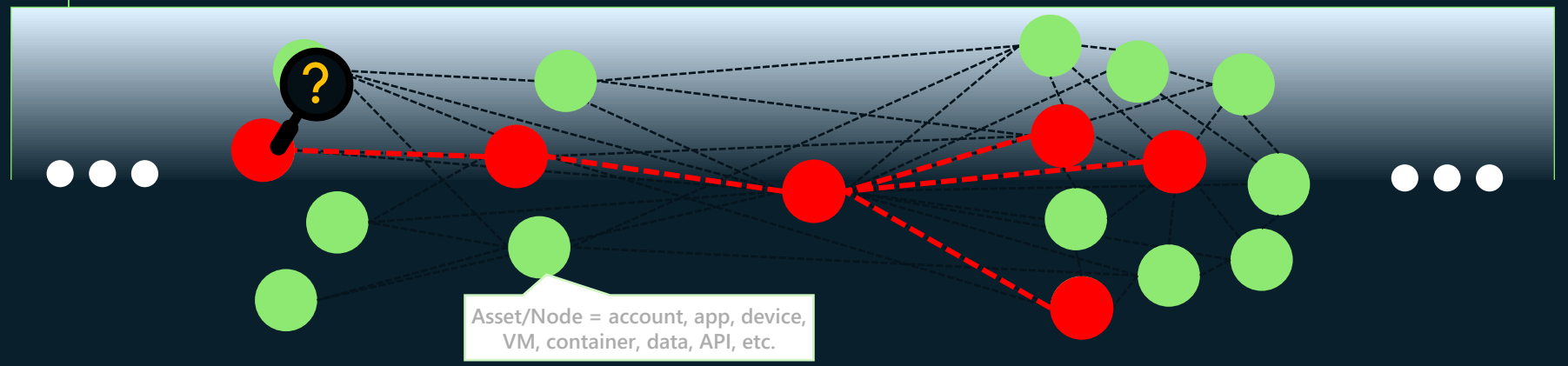
- Unify conditional access
- Ensure least privilege access
- Improve the user experience
- Reduce on-premises footprint

Zero Trust Principles

Business Enablement
Align security to the organization's mission, priorities, risks, and processes

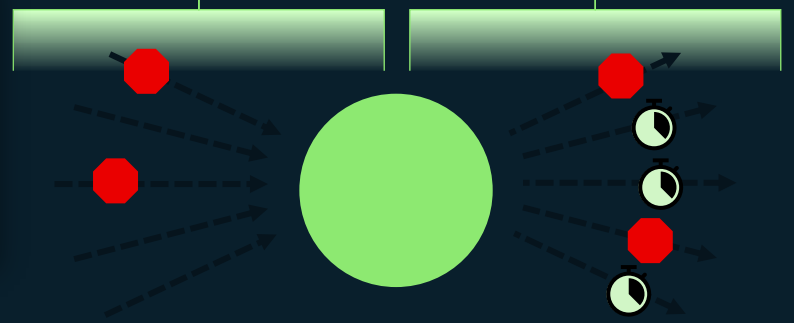
Assume Breach (Assume Compromise)
Assume attackers can and will successfully attack anything (identity, network, device, app, infrastructure, etc.) and plan accordingly

→ *Transforms from ~~"defend the network"~~ to "enable secure productivity on any network"*



Verify explicitly
Protect assets against attacker control by explicitly validating that all trust and security decisions use all relevant available information and telemetry.

→ *Reduces "attack surface" of each asset*

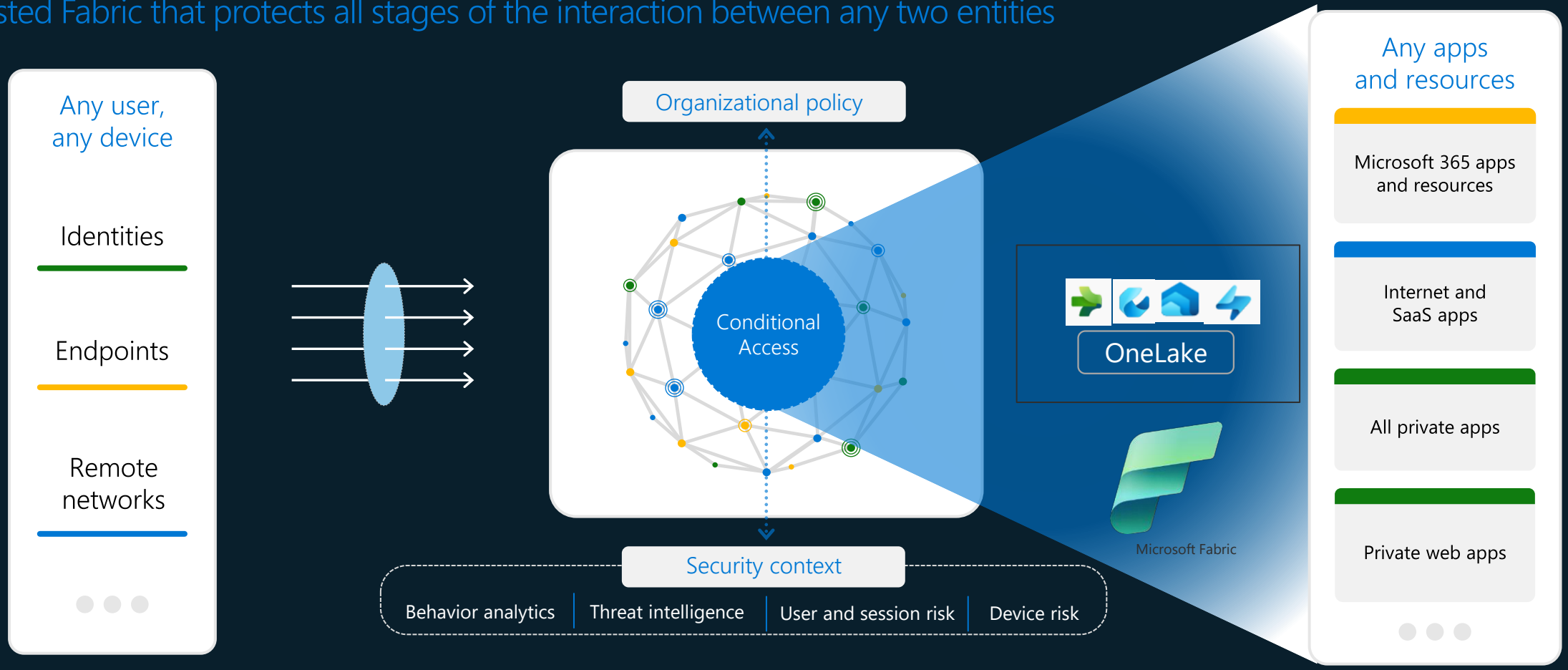


Use least privilege access
Limit access of a potentially compromised asset, typically with just-in-time and just-enough-access (JIT/JEA) and risk-based polices like adaptive access control.

→ *Reduce "blast radius" of compromises*

Conditional Access - Microsoft's Adaptive Access Vision

Trusted Fabric that protects all stages of the interaction between any two entities



Centralized control

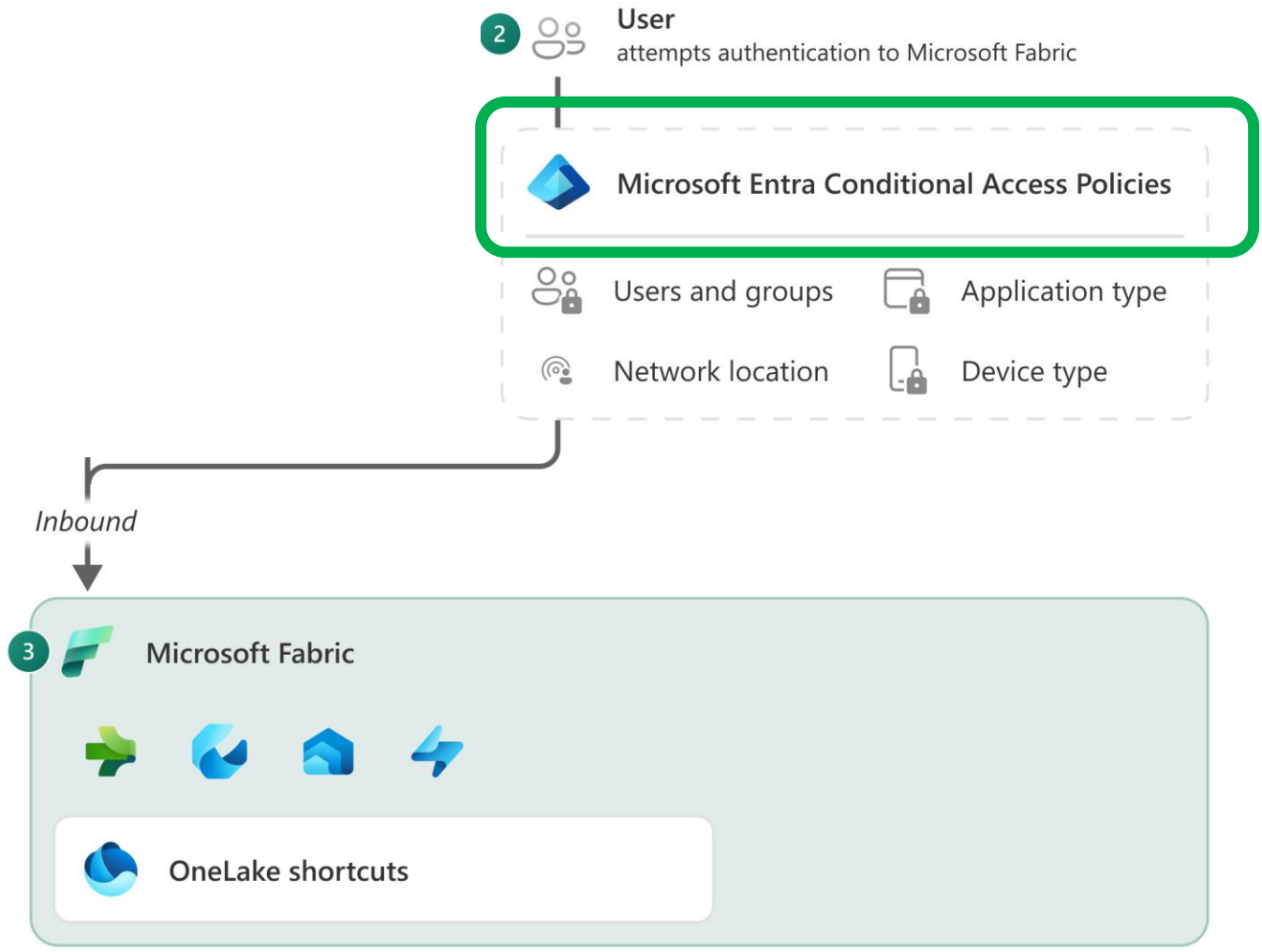
Unified Zero Trust architecture and policy engine simplifies management of access controls and technologies (Directory, SSO, Federation, RBAC, proxy, and more)

Consistent enforcement

Centralized policy is consistently applied across all resources where the action happens (identity, data, network + infra and apps across cloud, on-premises, IoT, OT, and more)

Microsoft Fabric Conditional Access

- Conditional access requires Microsoft Entra ID P1 licenses.




DEMO

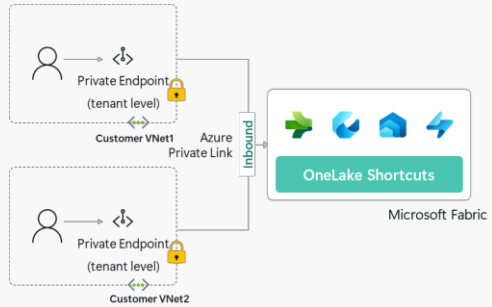
Inbound and Outbound



- Regulations & Certification
- Data Encryption
- E2E Auditability
- High Availability & Disaster Recovery
- Microsoft Purview Information Protection
- Advanced tools in Microsoft Purview
- 3rd party and in-house governance & security solutions

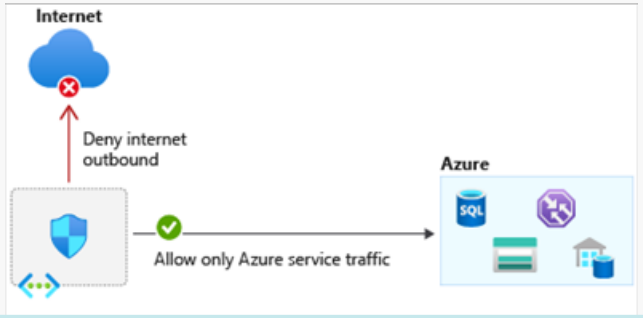
Secure inbound and outbound network connections

Secure and protect data across your organization



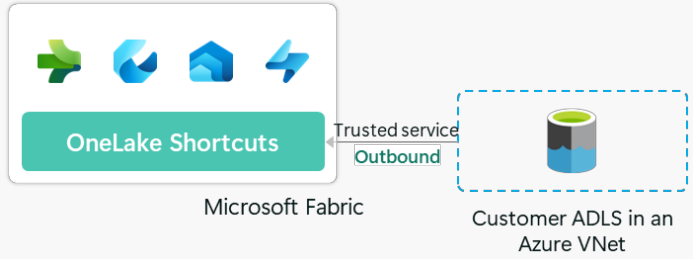
Fabric inbound security

Seamlessly secure inbound data access scenarios with Fabric's holistic identity and network security



Service tags

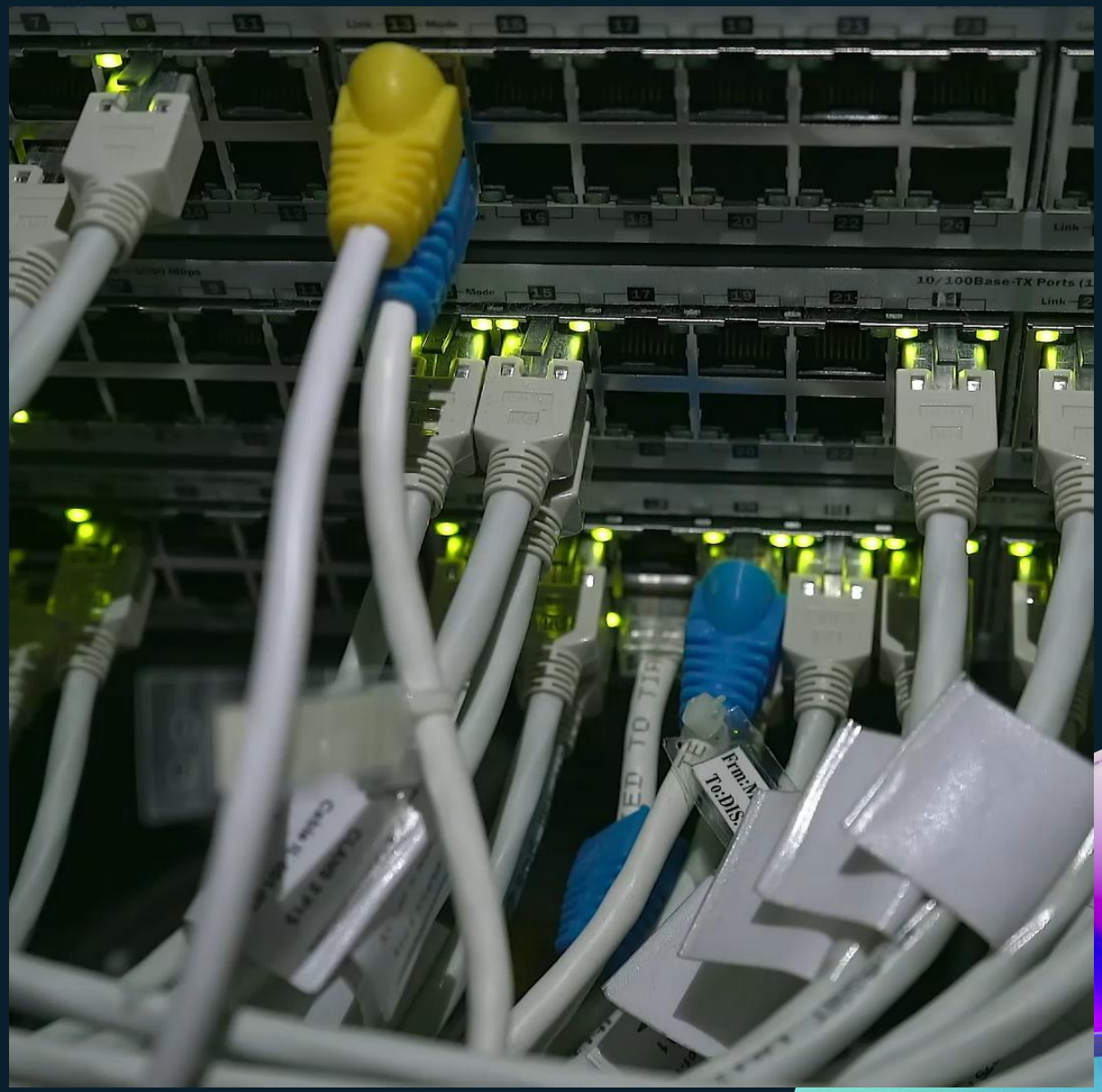
Minimize the complexity of updating network security rules using Azure service tags to group and manage IP addresses for a service



Fabric outbound security

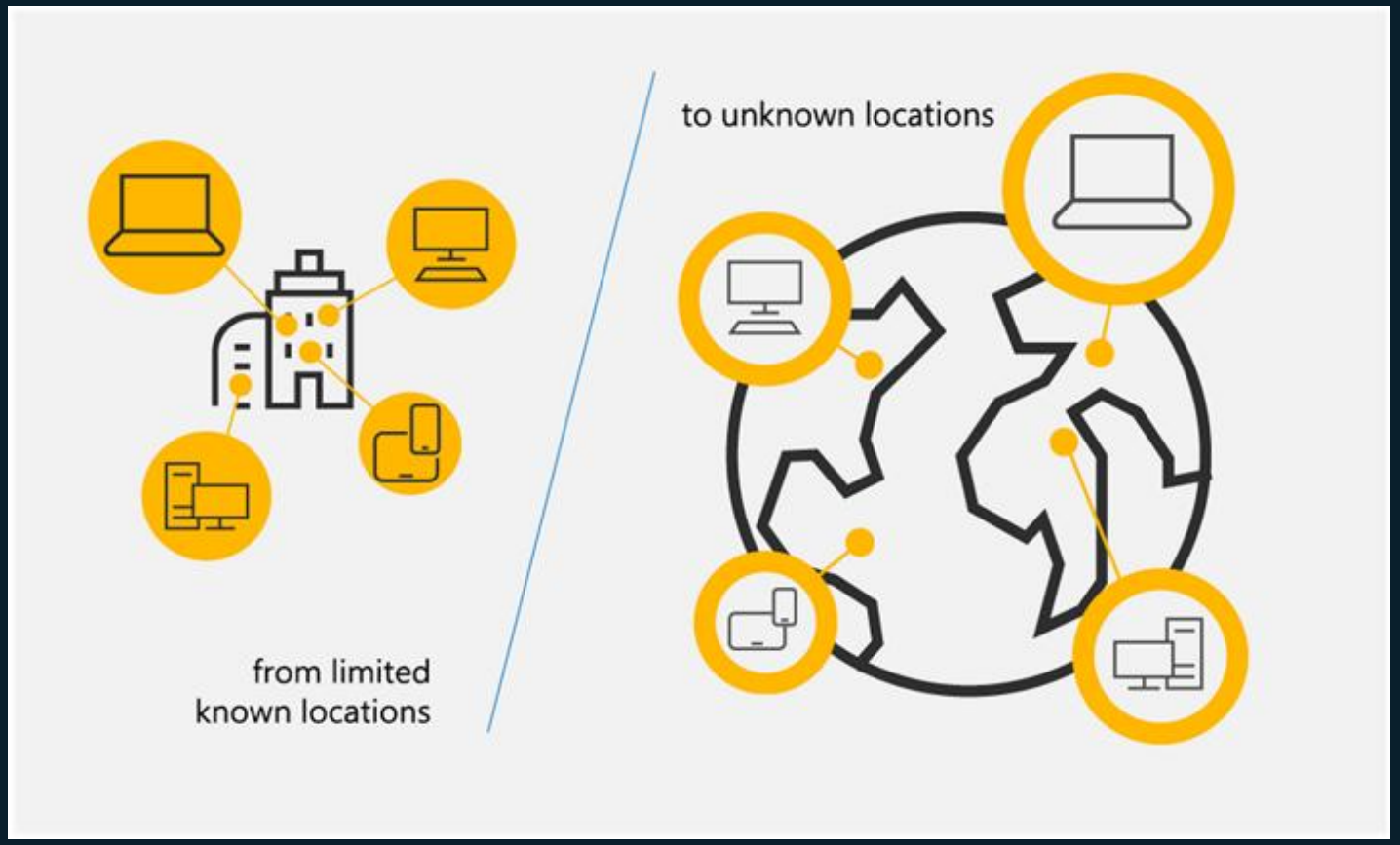
Securely connect to your protected data sources and/or ingest data into OneLake

Fabric Inbound Security



Inbound protection options

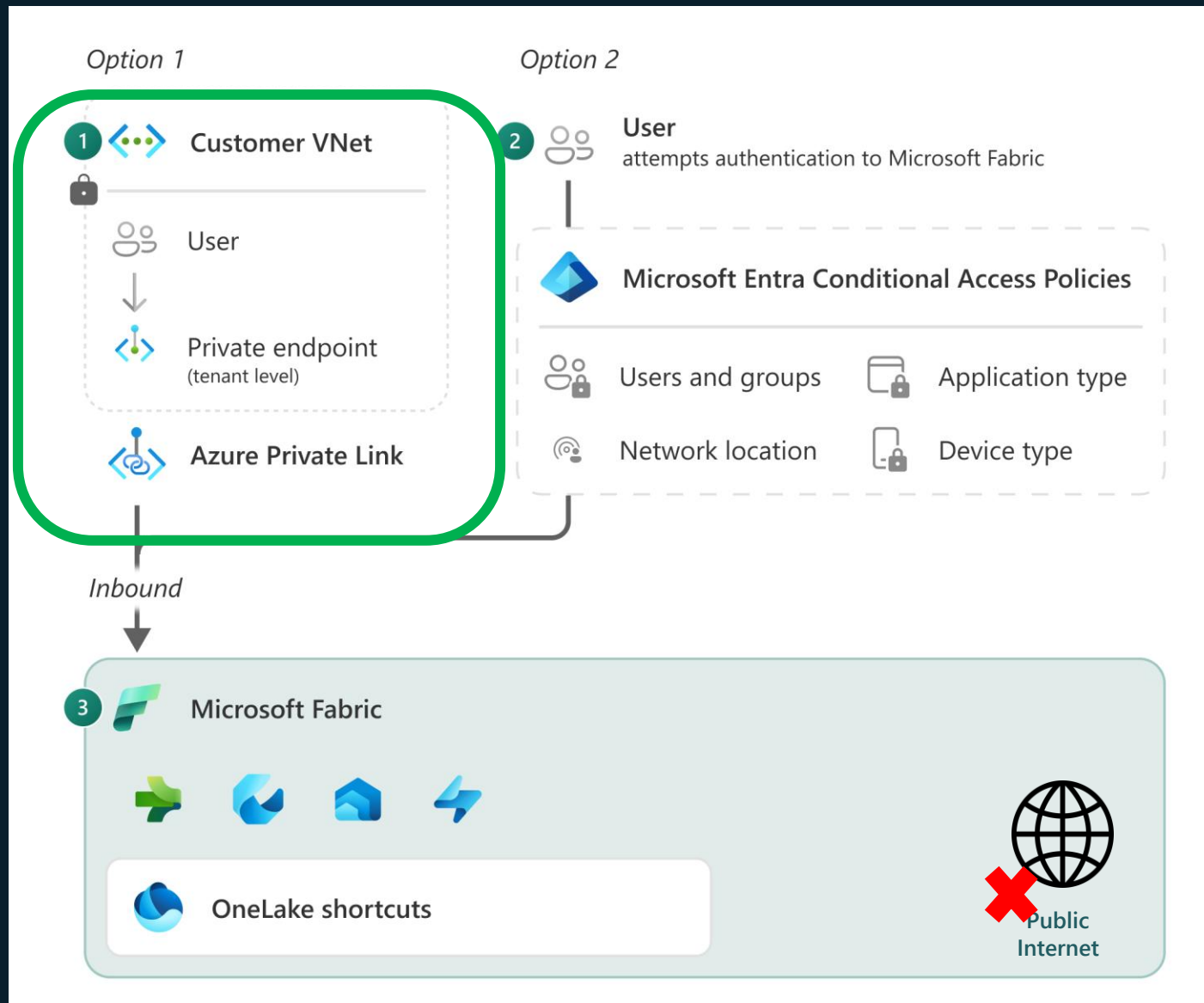
Perimeter Network Security **Zero Trust Approach(Conditional Access)**

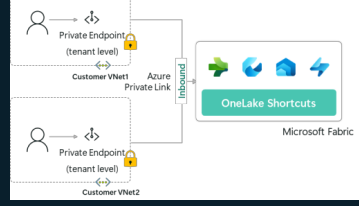


Private Link for Fabric

- What it means:

1. Fabric is disconnected from the public internet
2. Every users needs to connect to the private network to get access on every device
3. No longer able to load resources locally (slower reports)
4. Increases ExpressRoute bandwidth and added costs for Private Links
5. Several product limitations (like on-prem data gateway)
6. Only available on tenant level(workspace roadmap)

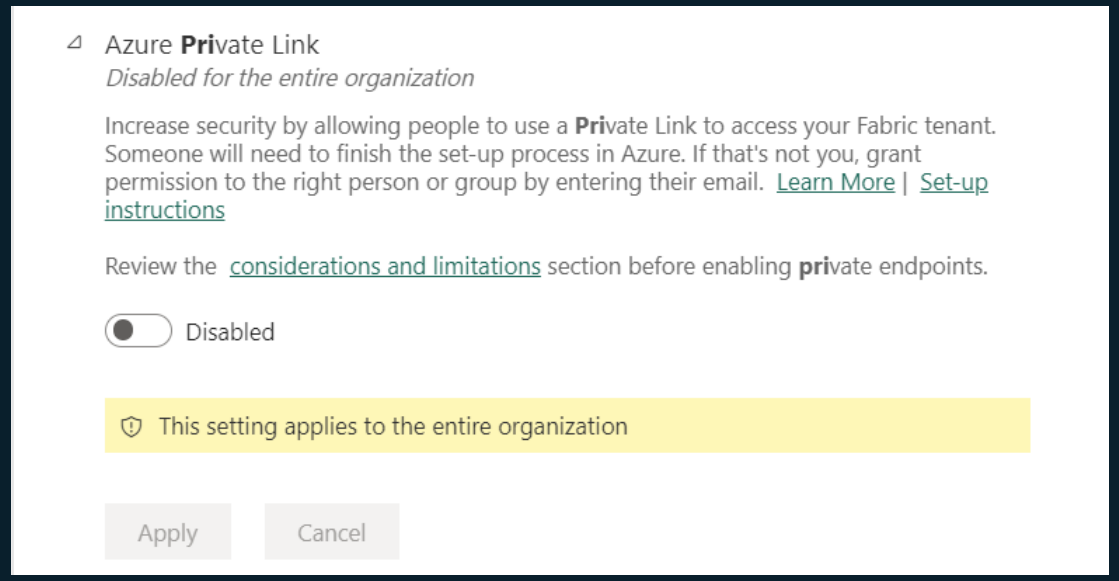


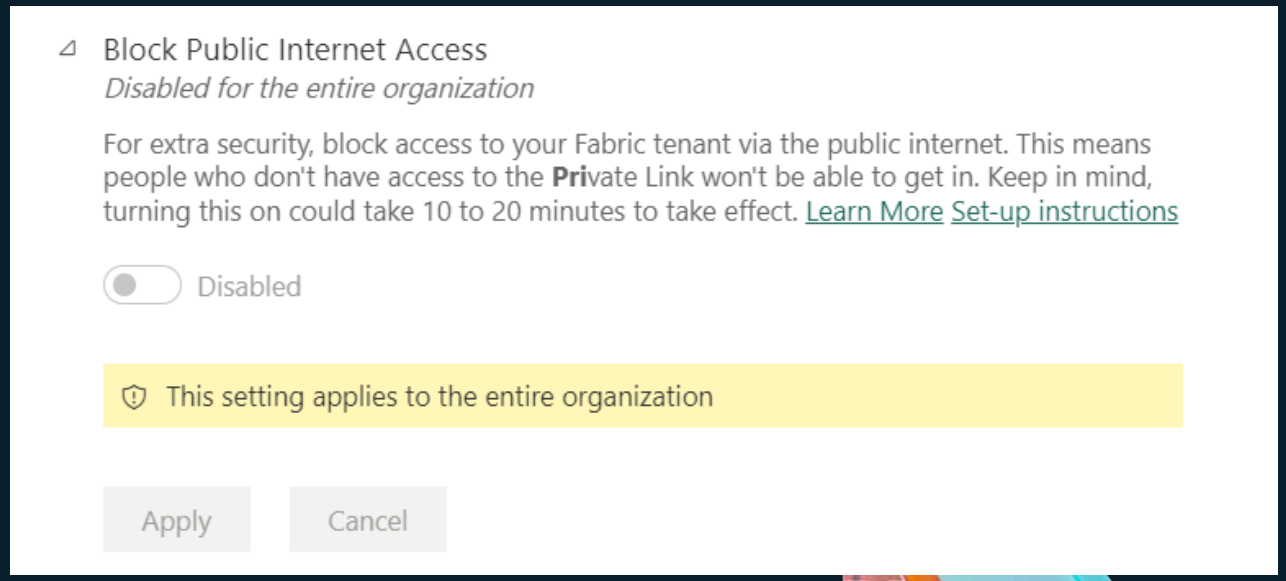


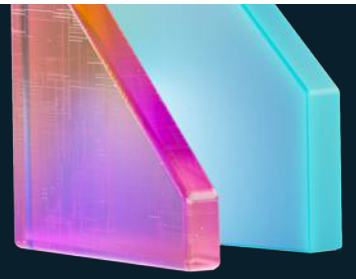
Private Link for Fabric

- Admin Portal







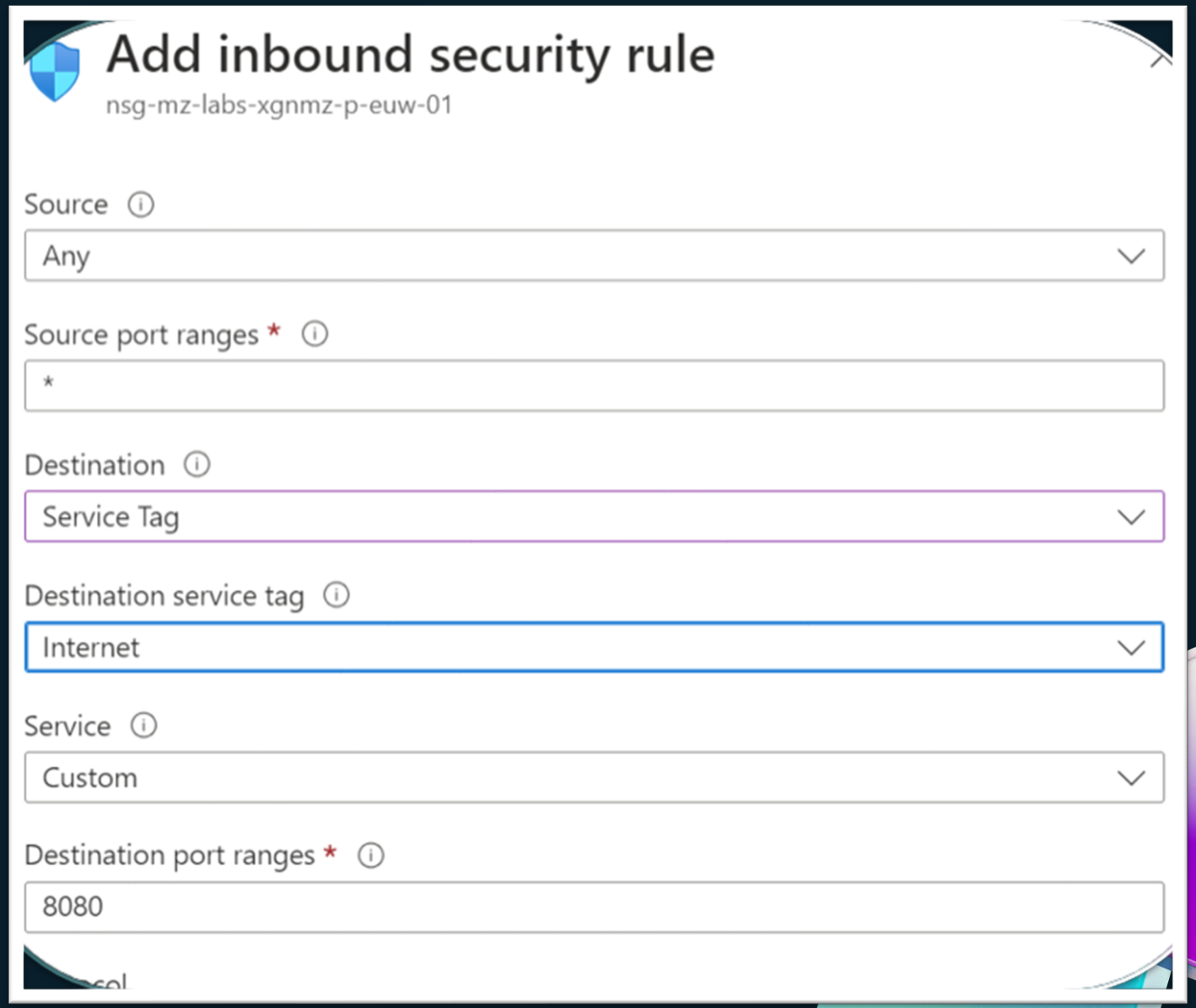


Private Links

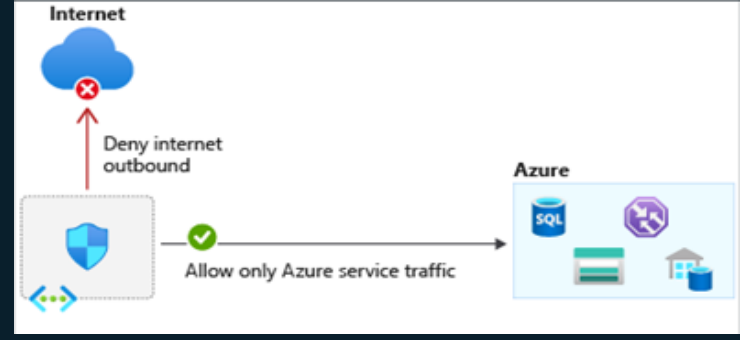
Limitations

- Fabric supports up to 450 capacities in a tenant where Private Link is enabled.
- When capacity is newly created, it won't support private link until its endpoint is reflected in the private DNS zone. This can take up to 24 hours. Trial capacity is not supported
- Tenant migration is blocked when Private Link is turned on in the Fabric admin portal.
- Customers can't connect to Fabric resources in multiple tenants from a single virtual network, but rather only the last tenant to set up Private Link.
- Each private endpoint can be connected to one tenant only. You can't set up a private link to be used by more than one tenant.
- On-premises data gateways aren't supported and fail to register when Private Link is enabled. To run the gateway configurator successfully, Private Link must be disabled. Learn more about this scenario. VNet data gateways will work.

Service Tags

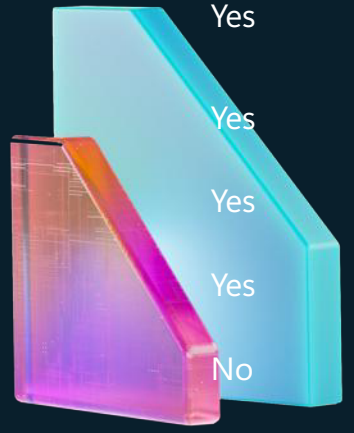


Service Tags



- You can use Azure service tags to enable connections to and from Microsoft Fabric.
- A service tag is a defined group of IP addresses that is automatically managed, as a group, to minimize the complexity of updates or changes to network security rules
- You can use the service tags to define network access controls:
  - Network security groups.
  - Azure Firewall.
  - User-defined routes..

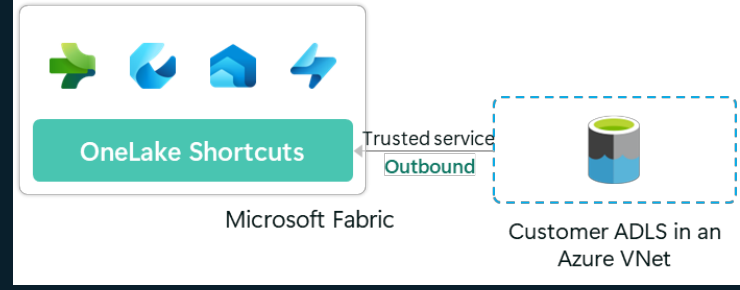
DataFactory	Azure Data Factory	Both	No	Yes
DataFactoryManagement	On premises data pipeline activity	Outbound	No	Yes
EventHub	Azure Event Hubs	Outbound	Yes	Yes
Power BI	Power BI and Microsoft Fabric	Both	No	Yes
PowerQueryOnline	Power Query Online	Both	No	Yes
KustoAnalytics	Real-Time Intelligence	Both	No	No
SQL	Fabric SQL database	Both	Yes	Yes



Fabric Outbound Security



Getting data into Fabric



Trusted Access for Fabric Workspace Identity

Storage Account (ADLS)

Workspace 1
Shortcuts

Workspace 2
Shortcuts

Workspace N

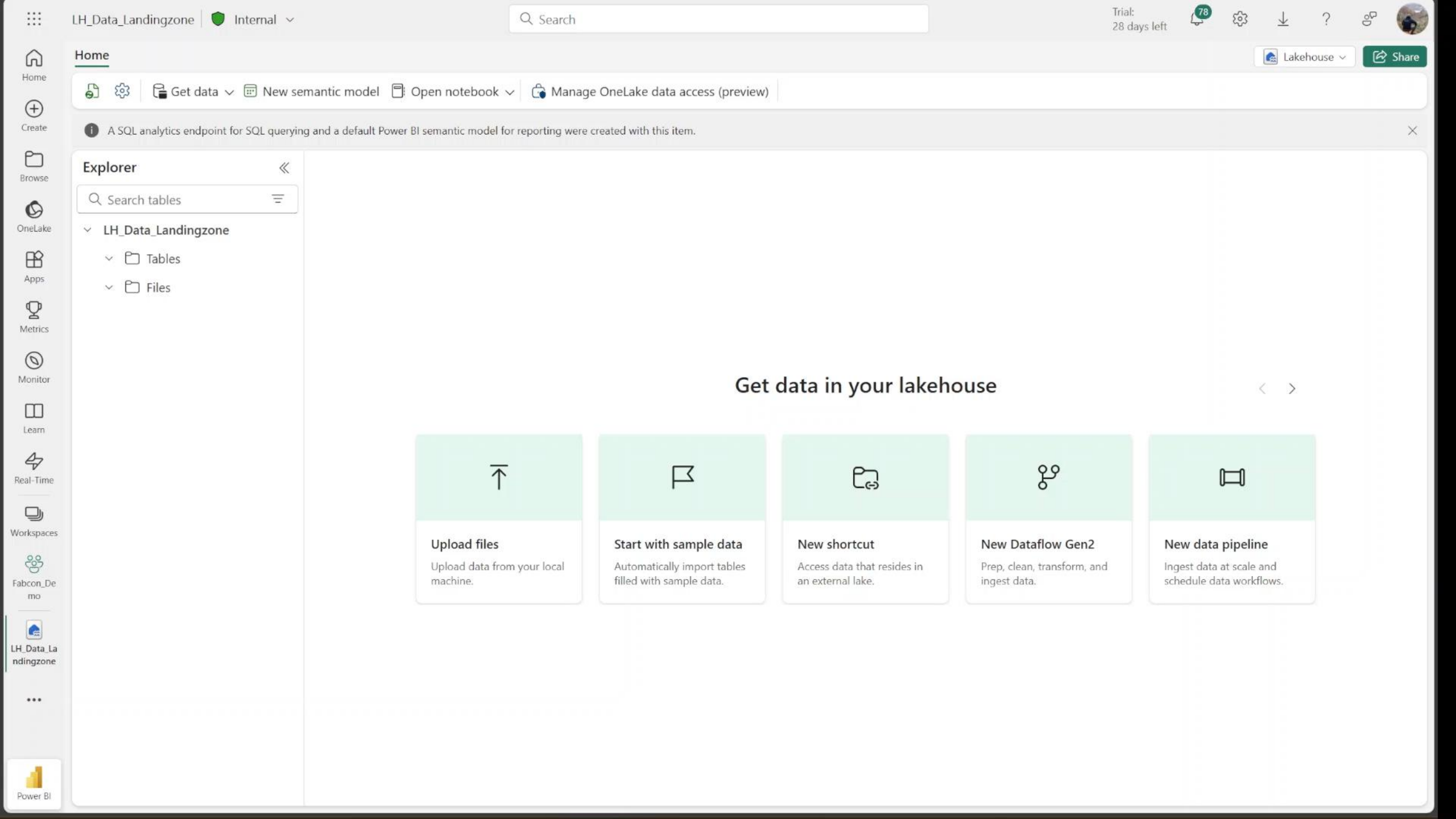
Managed Private Endpoint

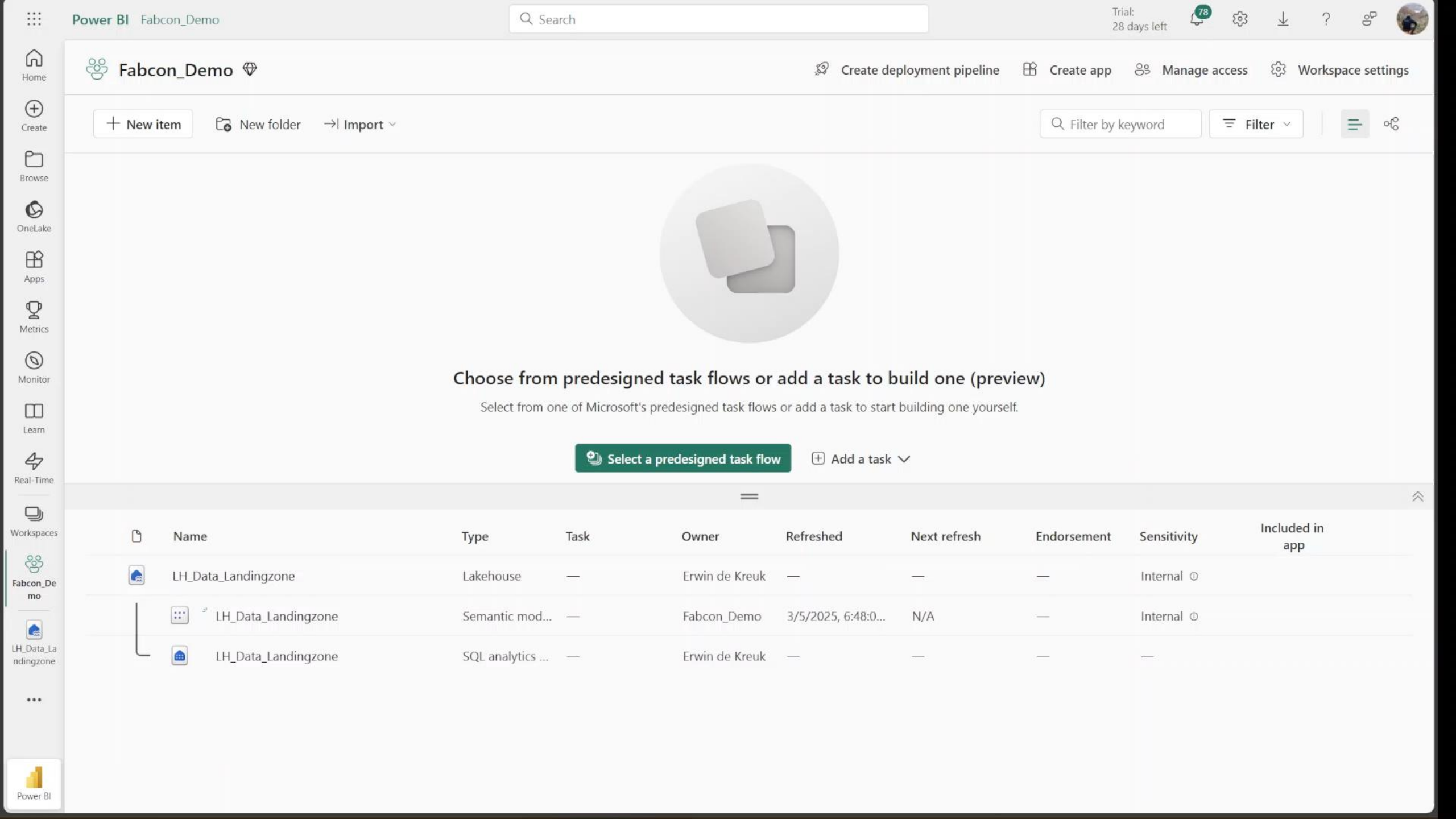
Customer VNet

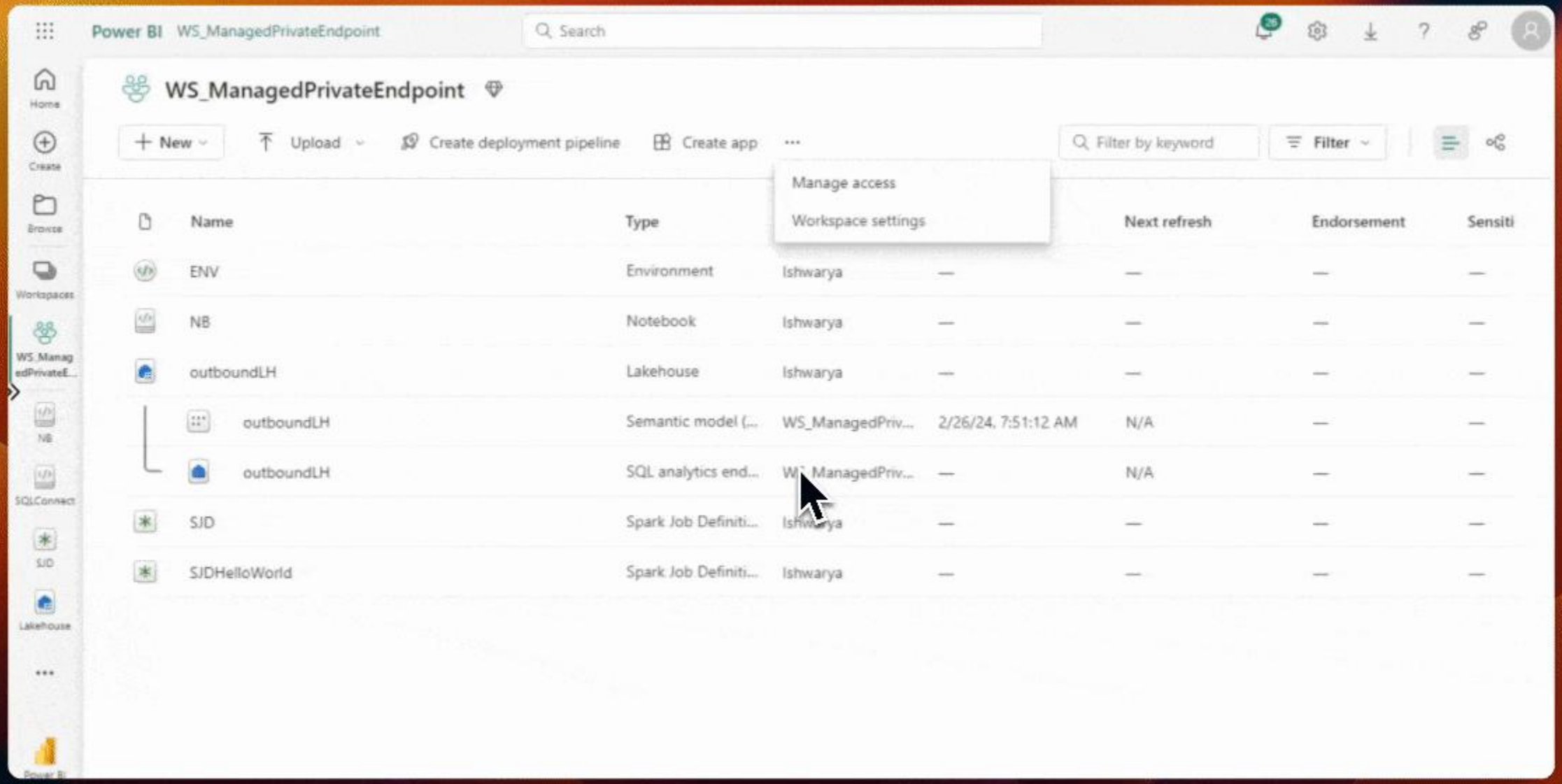
Public Internet

Create and use managed private endpoints in Microsoft Fabric - Microsoft Fabric | Microsoft Learn

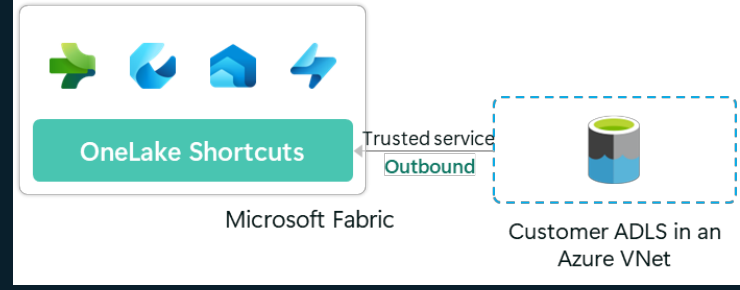

DEMO

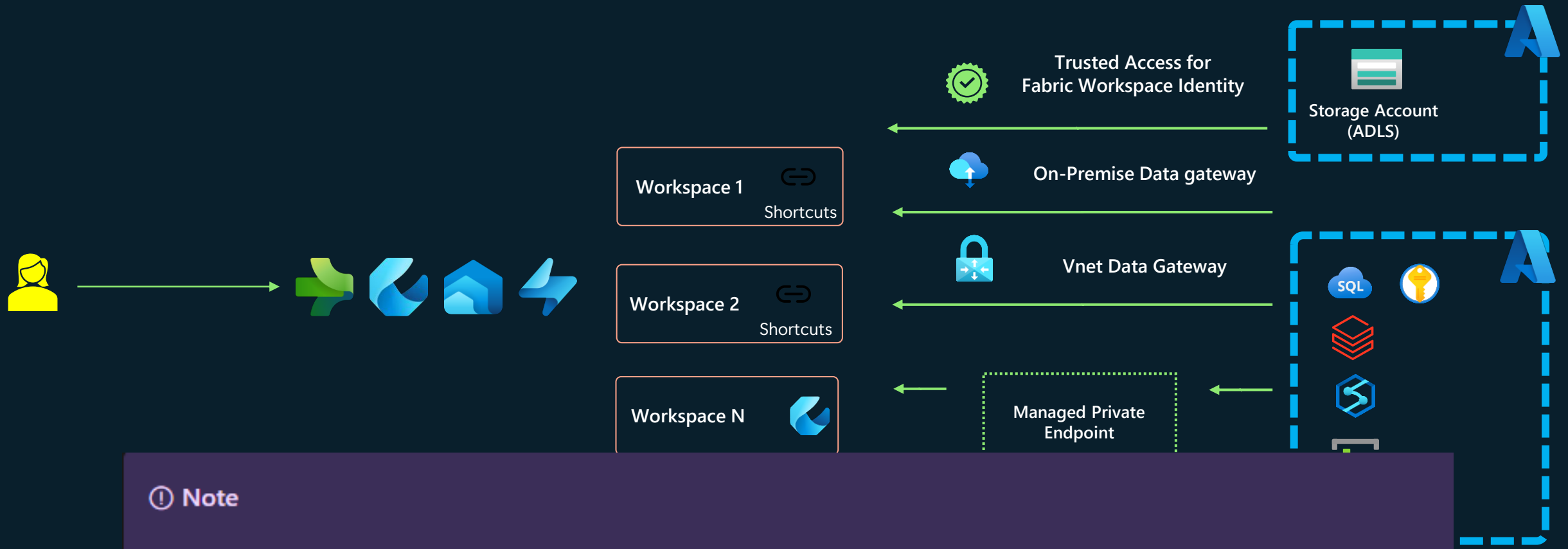

LH_Data_Landingzone
Internal
Search
Trial:
28 days left
78
Home
Lakehouse
Share
Get data
New semantic model
Open notebook
Manage OneLake data access (preview)
A SQL analytics endpoint for SQL querying and a default Power BI semantic model for reporting were created with this item.
Explorer
Search tables
LH_Data_Landingzone
Tables
Files
Get data in your lakehouse
Upload files
Upload data from your local machine.
Start with sample data
Automatically import tables filled with sample data.
New shortcut
Access data that resides in an external lake.
New Dataflow Gen2
Prep, clean, transform, and ingest data.
New data pipeline
Ingest data at scale and schedule data workflows.
Home
Create
Browse
OneLake
Apps
Metrics
Monitor
Learn
Real-Time
Workspaces
Fabcon_Demo
LH_Data_Landingzone
Power BI


Power BI
Fabcon_Demo
Search
Trial:
28 days left
78
Fabcon_Demo
Create deployment pipeline
Create app
Manage access
Workspace settings
New item
New folder
Import
Filter by keyword
Filter
Choose from predesigned task flows or add a task to build one (preview)
Select from one of Microsoft's predesigned task flows or add a task to start building one yourself.
Select a predesigned task flow
Add a task
Name
Type
Task
Owner
Refreshed
Next refresh
Endorsement
Sensitivity
Included in app
LH_Data_Landingzone
Lakehouse
Erwin de Kreuk
Internal
LH_Data_Landingzone
Semantic mod...
Fabcon_Demo
3/5/2025, 6:48:0...
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Internal
LH_Data_Landingzone
SQL analytics ...
Erwin de Kreuk
Home
Create
Browse
OneLake
Apps
Metrics
Monitor
Learn
Real-Time
Workspaces
Fabcon_Demo
LH_Data_Landingzone
Power BI


Power BI
WS_ManagedPrivateEndpoint
Search
WS_ManagedPrivateEndpoint
New
Upload
Create deployment pipeline
Create app
Filter by keyword
Filter
Manage access
Workspace settings
Name
Type
Next refresh
Endorsement
Sensiti
ENV
Environment
Ishwarya
NB
Notebook
Ishwarya
outboundLH
Lakehouse
Ishwarya
outboundLH
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WS_ManagedPriv...
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outboundLH
SQL analytics end...
N/A
SJD
Spark Job Definiti...
Ishwarya
SJDHelloWorld
Spark Job Definiti...
Ishwarya
Home
Create
Browse
Workspaces
NB
SQLConnect
SJD
Lakehouse

Getting data into Fabric





Note

Trusted workspace access is **generally available**, but can only be used in F SKU capacities. For information about buying a Fabric subscription, see Buy a Microsoft Fabric subscription. Trusted workspace access is not supported in Trial capacities.

Create and use managed private endpoints in Microsoft Fabric - Microsoft Fabric | Microsoft Learn

Microsoft Purview



Reshape how you access, manage, and act on data and insights by connecting every data source and analytics service together in Fabric

Keep your data safe and governed with unified data governance, information protection, and risk and compliance solutions

Data ownership	Unmatched security & compliance	Unified governance

Microsoft Purview

Data Security



For information and cybersecurity teams

Data Loss Prevention
Insider Risk Management
Information Protection

Data Governance



For data consumers, data owners, data officers

Data Discovery
Data Curation & Management
Data Estate Health

Risk & Compliance



For risk, compliance, and legal teams

Data Lifecycle Management
eDiscovery & Audit
Communication Compliance

On-prem and multi-cloud

Unstructured & structured data

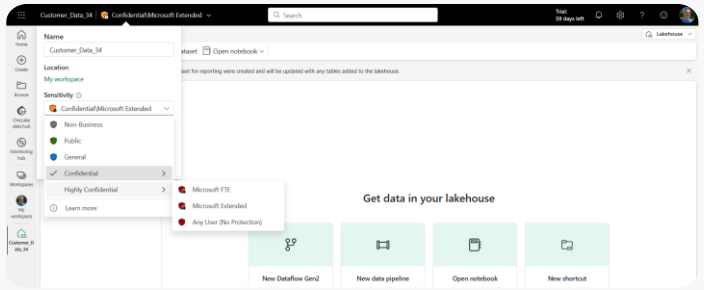
Across IaaS, and SaaS

Microsoft Purview

Seamlessly secure inbound data access scenarios with Fabric's holistic identity and network security

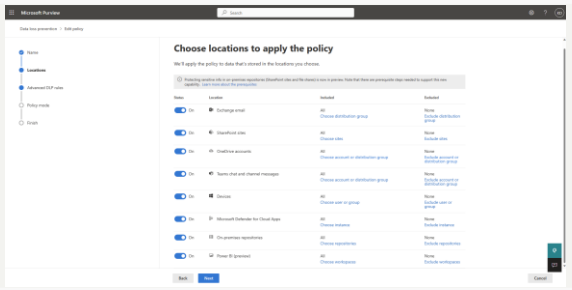
*Additional Microsoft Purview purchase required

Secure and protect data across your organization



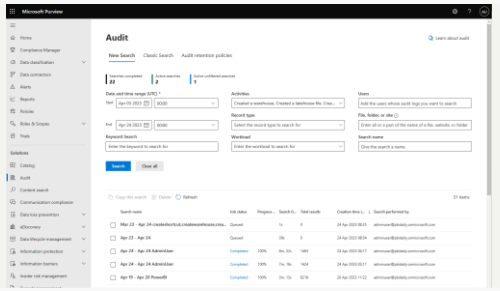
Information Protection labels

Classify sensitive Fabric data using the same sensitivity labels that are used in Microsoft 365—enforced even when the data is exported



Data Loss Prevention policies

Automatically detect the upload of sensitive data such as PII and trigger automatic risk remediation actions such as alerts



Audit

Log user activities from Microsoft Fabric
to support security, forensic, and internal investigations

Microsoft Fabric Compliancy

- GDPR
- EUDB
- ISO certifications including ISO 27001, 27701, 27017, 27018
- HIPAA, HITRUST compliant
- SOX compliant
- SOC 1, SOC 2, Type 2
- FedRamp



Roadmap

- **Private Link support at a workspace level**
- **Data exfiltration protection for Spark (Sprink)**
- **Workspace Public IP Firewall**
- **Customer Managed Keys**



More sessions

- **Building a Metadata-Driven Framework with SQL database in Fabric and Data Factory**
  - Grand Ballroom 120
  - Wednesday 14:00 – 15:00
  - Erwin de Kreuk

- **Securing Your Fabric Workloads - Best Practices from the Fabric Security Team**
  - **Wednesday 9:15 AM - 10:15 AM Boulevard Ballroom 162**

- **Deep Dive on OneLake Data Security**
  - **Wednesday 10:00 AM - 11:00 AM Premier Ballroom 313 Aaron Merrill and Cristian Petculescu**

- **Network Security for SaaS Platform**
  - **Tuesday 3:15 PM - 4:15 PM Grand Ballroom 119 Rick Xu and Arthi Ramasubramanian Iyer**

- **Microsoft Purview Data Security in the Era of AI**
  - **Wednesday 10:00 AM - 11:00 AM Grand Ballroom 120 Maithili Dandige and Joanne Marone**



Let's connect

Erwin de Kreuk

- Principal Consultant
- Lead Data & AI InSpark

@erwindekreuk.bsky.social

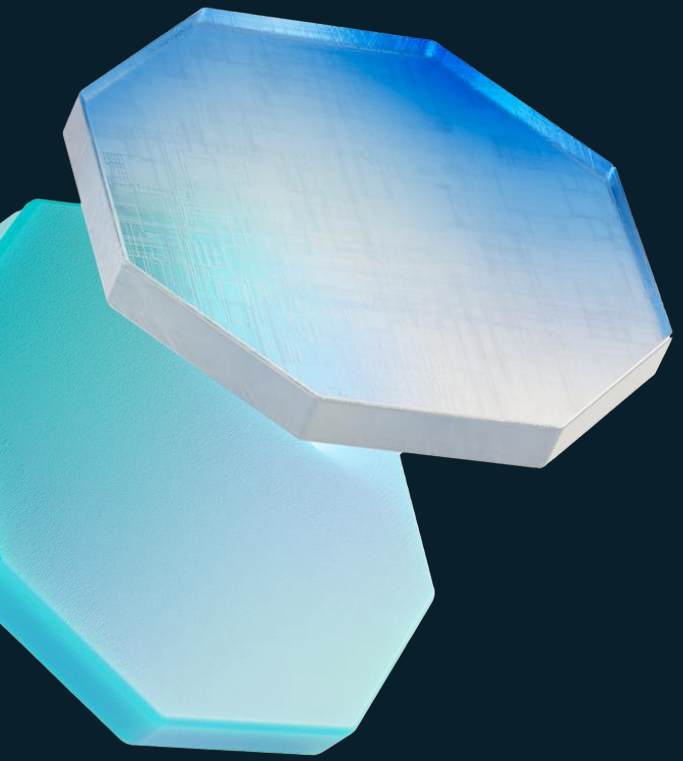
linkedin.com/in/erwindekreuk

erwindekreuk.com

github.com/edkreuk

https://sessionize.com/erwin-de-kreuk/

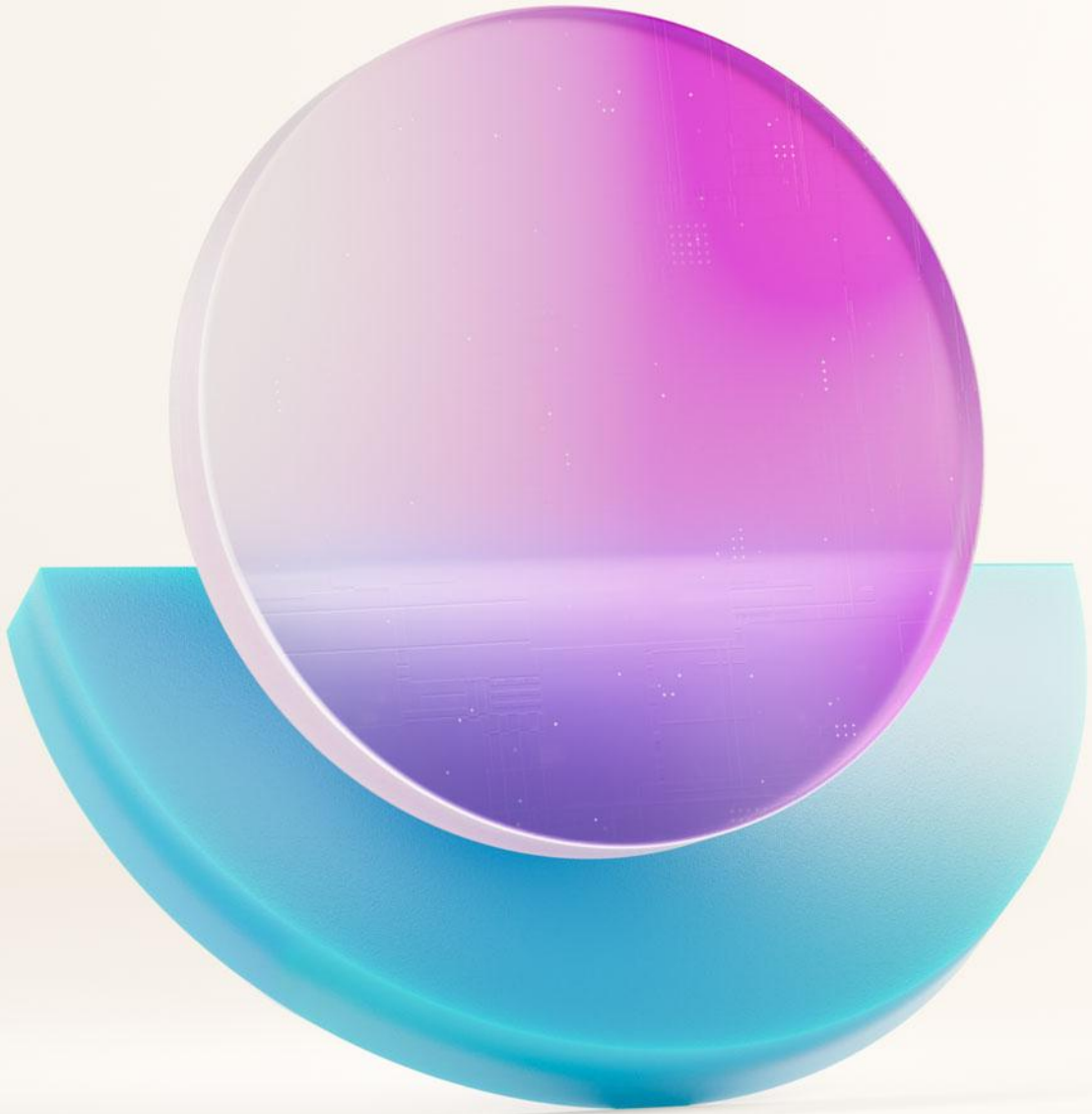
Dutchfabricusergroup.com





Microsoft® MVP Most Valuable Professional

Microsoft Fabric
Community Conference



Fast-track your career in data and AI

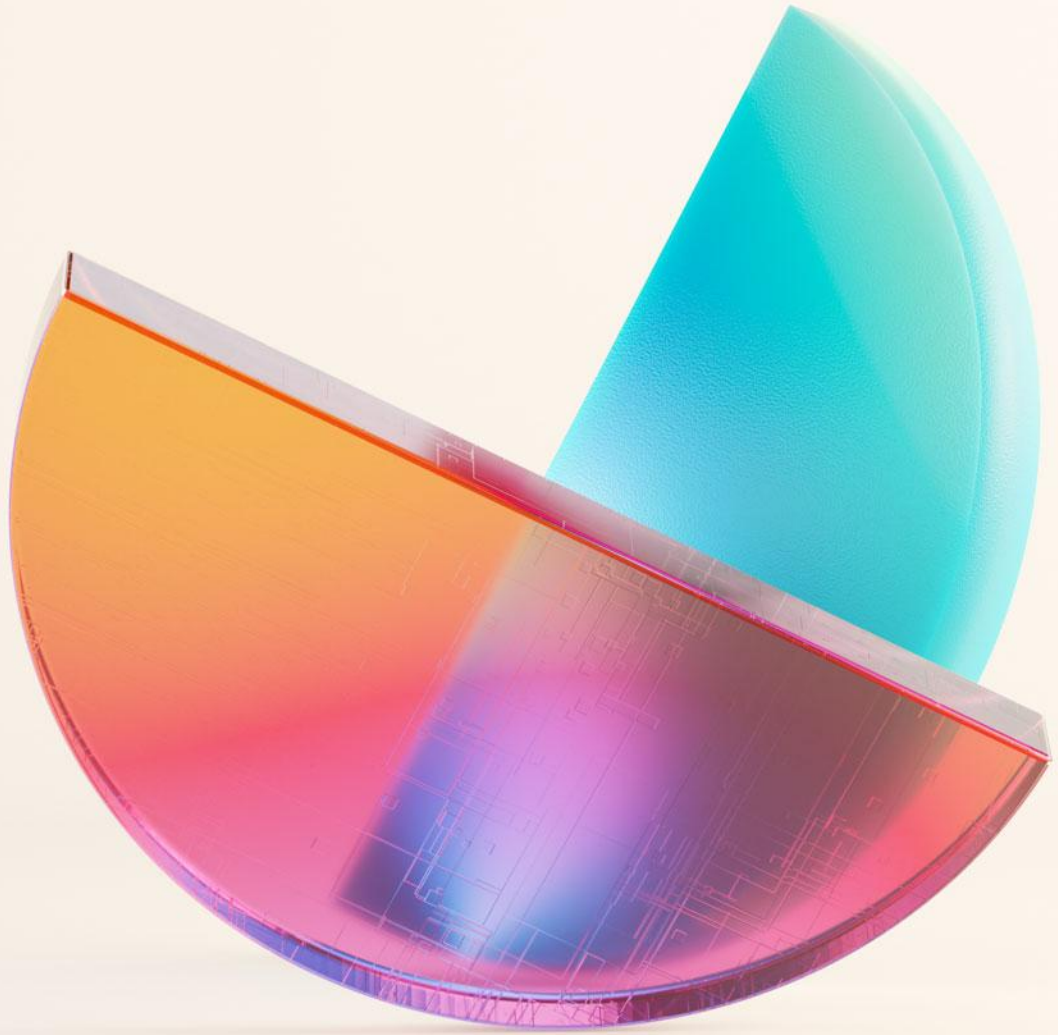
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aka.ms/fabcon/dp700

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